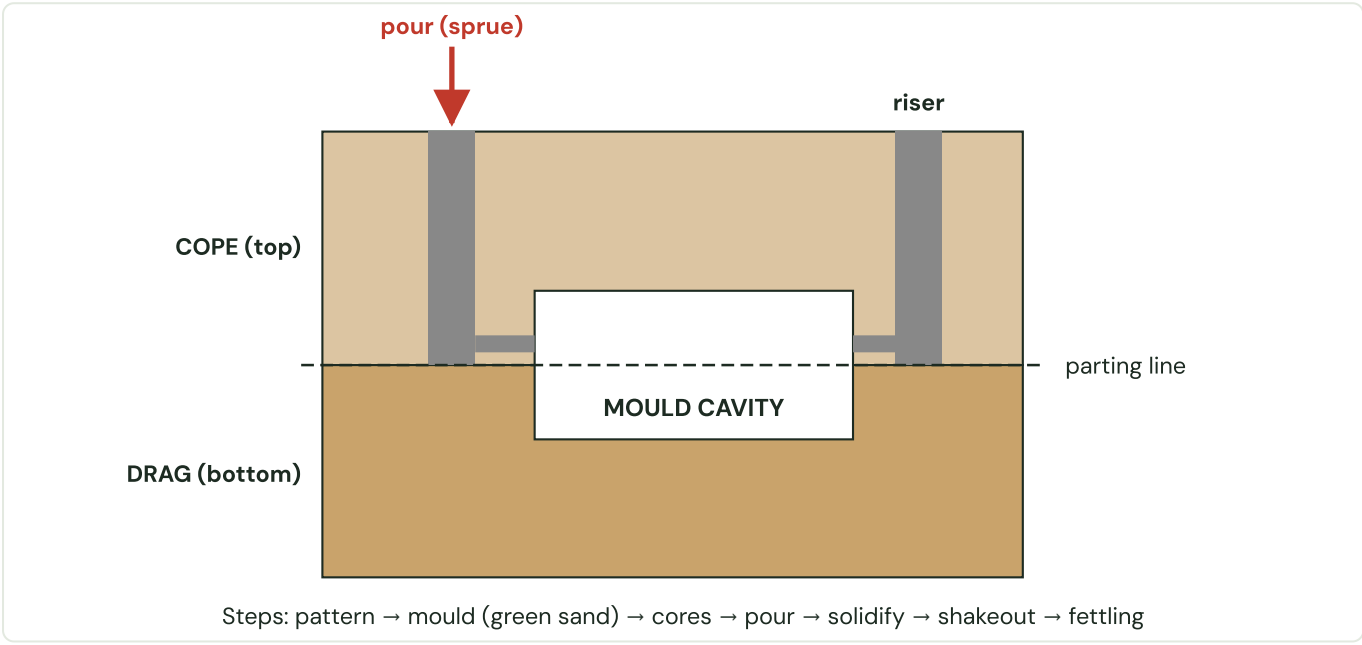
UNIT 8 · TOPIC 5

Casting and Die-Casting Processes

Syllabus topic: Casting and die-casting processes

Sand casting

Casting = pouring molten metal into a mould cavity, letting it solidify, and removing the casting. SAND CASTING is the most common and cheapest, and works for any metal including ferrous.



Sand casting mould (redrawn — the source figure's pouring arrow pointed out of the sprue)

Element	Role
PATTERN (wood/metal/plastic)	Forms the cavity in green (moist) sand
Flask — COPE (top) and DRAG (bottom)	Two-part box split at the PARTING line
SPRUE / gating system	Molten metal enters here
RISER	Feeds metal to compensate shrinkage
CORES	Form internal cavities

Steps:

1. Pattern-making.
2. Moulding.
3. Core-setting.
4. Pouring.
5. Solidification.
6. Shakeout.
7. Fettling (cleaning).

Pattern allowances: shrinkage, machining, DRAFT (taper for easy withdrawal), distortion and shake.

Die casting

- Molten NON-ferrous metal (Al, Zn, Mg) is forced under HIGH PRESSURE into a permanent hardened-steel DIE.
- High accuracy, fine finish, high production rate (thousands of cycles) — but high tooling cost, so it suits MASS production of small parts.
- **Hot-chamber** machines for Zn (low-melting); **cold-chamber** for Al (higher-melting).

	Sand casting	Die casting
Mould	Expendable sand (one use)	Permanent steel die (1000s of cycles)
Metals	Any, including ferrous	Mainly non-ferrous (Al, Zn, Mg)
Pressure	Gravity pour	High-pressure injection
Size	Large and small	Small–medium
Accuracy / finish	Lower (machining needed)	High, near-net shape
Cost	Low tooling — prototypes	High tooling — mass production

Other casting processes

- Permanent-mould (gravity die) casting.
- **INVESTMENT (lost-wax)** casting — intricate precision parts.
- Centrifugal casting — pipes, liners.
- Shell moulding; continuous casting.

Casting defects

Defect	Cause
Blowholes / porosity	Trapped gas
Shrinkage cavity	Inadequate feeding
Misrun / cold shut	Incomplete filling
Inclusions	Foreign material
Hot tears	Restrained contraction
Mismatch	Cope/drag misalignment

Practice MCQs

Q71. In sand casting, the top half of the moulding flask is the:

(a) Drag (b) Cope
(c) Core (d) Riser

Q72. The bottom half of the flask is the:

(a) Cope (b) Drag
(c) Sprue (d) Gate

Q73. Molten metal is poured into the mould through the:

(a) Riser
(b) Sprue/gating system
(c) Core
(d) Vent

Q74. A riser in a mould serves to:

(a) Vent gases only
(b) Feed metal to compensate shrinkage
(c) Locate the core
(d) Cool the metal

Q75. Internal cavities in a casting are produced using a:

(a) Riser
(b) Core
(c) Chill
(d) Sprue

Q76. Which is NOT a pattern allowance?

(a) Shrinkage
(b) Draft
(c) Machining
(d) Porosity

Q77. Draft allowance is provided to:

(a) Add strength
(b) Allow easy withdrawal of the pattern
(c) Increase weight
(d) Cool faster

Q78. Die casting is generally limited to:

(a) Ferrous metals
(b) Non-ferrous metals (Al, Zn, Mg)
(c) Ceramics
(d) Wood

Q79. Compared with sand casting, die casting gives:

(a) Lower accuracy
(b) Higher accuracy and finish, high production rate
(c) Cheaper tooling
(d) Larger castings only

Q80. Sand casting is preferred over die casting for:

(a) Mass production of small Al parts
(b) Large parts, ferrous metals, prototypes (low tooling cost)
(c) Zinc toys
(d) Thin wall precision

Q81. Lost-wax precision casting is also called:

(a) Shell moulding
(b) Investment casting
(c) Centrifugal casting
(d) Die casting

Q82. A casting defect from trapped gas is:

(a) Hot tear
(b) Blowhole/porosity
(c) Misrun
(d) Mismatch

Q83. Incomplete filling of the mould cavity gives:

(a) Shrinkage cavity
(b) Misrun/cold shut
(c) Inclusion
(d) Warping

Answers

Q71 — (b) Cope — *Sand-casting anatomy — cope is the top half.*
Q72 — (b) Drag — *Drag is the bottom half.*
Q73 — (b) Sprue/gating system
Q74 — (b) Feed metal to compensate shrinkage — *Riser feeds shrinkage.*
Q75 — (b) Core — *Core forms internal cavities.*
Q76 — (d) Porosity — *Porosity is a DEFECT, not an allowance.*
Q77 — (b) Allow easy withdrawal of the pattern — *Draft allows easy pattern withdrawal.*
Q78 — (b) Non-ferrous metals (Al, Zn, Mg) — *Die casting = permanent steel die, high pressure, non-ferrous.*
Q79 — (b) Higher accuracy and finish, high production rate — *Mass production with high accuracy.*

Q80 — (b) Large parts, ferrous metals, prototypes (low tooling cost) — *Sand casting = expendable mould, any metal, large parts, cheap tooling.*

Q81 — (b) Investment casting

Q82 — (b) Blowhole/porosity

Q83 — (b) Misrun/cold shut

Sources: Hajra Choudhury — Elements of Workshop Technology (Vol I: casting, welding); Kalpakjian — Manufacturing Engineering; AWS welding references; literature on HAZ, weld defects, NDT and sand vs die casting (as given in the source notes, July 2026); NPTEL manufacturing lectures.